Tittle of the Project:- NUTRI LENS AI

ABSTRACT

PROBLEM STATEMENT:- Many consumers struggle to understand the ingredient lists on packaged food products due to complex scientific names and food additives as a result they often find it difficult to determine whether the product is suitable for their health , dietary, preference and fitness goals. Existing nutrition labels provide general information but do on offer personalized guidance based on factors such as age , allergies or medical conditions.

SOLUTION:- The proposed solution is an AI-powered application that helps users understand the nutritional quality of packaged food products in a simple and personalized way. Users can upload an image of a product's ingredient list, and the system uses Optical Character Recognition to extract the text.

It then analyses the ingredients using AI and a nutrition database to identify harmful additives, allergens, preservatives, and beneficial ingredients. Based on the user's age, dietary preferences, fitness goals, allergies, and medical conditions, the application generates a personalized health rating and explains the reasons behind it in easy-to-understand language.

It also recommends healthier alternative products with better nutritional value, helping users compare options before making a purchase. By turning complex ingredient labels into clear and meaningful insights, the system enables consumers to make informed food choices and encourages healthier eating habits in their daily lives.

Software Requirements:

- Operating System: Windows 10/11, Linux, or macOS
- Programming Language:
- FRONTEND : HTML5, CSS3, JavaScript (ES6+)
- BACKEND : PYTHON (FAST API)
- Libraries/Frameworks: ReactJS (using Create React App / Vite)
- IDE/Development Tools: Visual Studio Code, Git for version control

Database: MY SQL

Hardware Requirements:

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB (8 GB recommended for faster processing)
- Storage: At least 500 GB HDD or 128 GB SSD
- Network: Required for API calls, updates, and hosting

Peripherals :

- Keyboard & mouse
- Internet connection

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